Hydrocarbons and Naming Alkanes

Answer key with worked solutions

A. Understanding

1. Definitions.

> (a) A compound of **carbon and hydrogen only**. (b) Containing **only single C–C bonds**, so each carbon holds the maximum number of hydrogen atoms (alkanes). (c) A family of compounds with the **same general formula** and similar chemistry, each member differing from the next by CH_2.

B. Formulas

2. Molecular formulas.

> (a) C_6H_{14} (b) $C_{10}H_{22}$ (c) C_4H_8 (d) C_3H_4 (e) $2n + 2 = 20$, so $n = 9$: C_9H_{20}

C. Naming

3. Names.

> (a) **butane** (b) **2-methylpropane** (c) **2-methylpentane** (d) **2,2-dimethylbutane** (e) **2,3-dimethylbutane** (f) **3-ethylpentane**

4. Condensed formulas.

> (a) $CH_3CH_2CH(CH_3)CH_2CH_2CH_3$ (b) $(CH_3)_3CCH_2CH(CH_3)_2$ (c) $CH_3CH(CH_3)CH(C_2H_5)CH_2CH_3$ (d) $CH_3CH{=}CHCH_3$

5. Corrected names.

> (a) **pentane**: a methyl on carbon 1 just extends the chain (longest chain missed). (b) **2-methylbutane**: the ethyl group is part of a longer chain (longest chain missed). (c) **2-methylbutane**: numbered from the wrong end (lowest locants). (d) **2,2-dimethylbutane**: numbered from the wrong end.

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D. Isomers and alkenes

6. Isomers of C_6H_{14}.

> **hexane** $CH_3CH_2CH_2CH_2CH_2CH_3$; **2-methylpentane** $CH_3CH(CH_3)CH_2CH_2CH_3$; **3-methylpentane** $CH_3CH_2CH(CH_3)CH_2CH_3$; **2,2-dimethylbutane** $CH_3C(CH_3)_2CH_2CH_3$; **2,3-dimethylbutane** $CH_3CH(CH_3)CH(CH_3)CH_3$

7. Alkenes.

> (a) **but-1-ene**, **but-2-ene**, **pent-1-ene**. (b) The chain must be numbered so the double bond gets the lowest locant; numbered from the other end, “but-3-ene” is **but-1-ene**.

E. Combustion

8. Burning octane.

> (a) $2\,C_8H_{18} + 25\,O_2 \longrightarrow 16\,CO_2 + 18\,H_2O$
>
> (b) $n(C_8H_{18}) = \dfrac{100.0\ \text{g}}{114.22\ \text{g/mol}} = 0.8755\ \text{mol}$
>
> $n(CO_2) = 0.8755\ \text{mol}\ C_8H_{18} \times \dfrac{16\ \text{mol}\ CO_2}{2\ \text{mol}\ C_8H_{18}} = 7.004\ \text{mol}$
>
> $m(CO_2) = 7.004\ \text{mol} \times 44.01\ \text{g/mol} = \mathbf{308.2\ g}$

F. Challenge

9. Formula from combustion data.

> (a) All the C ends up in CO_2 and all the H in H_2O:
>
> $n(\text{C}) = n(CO_2) = \dfrac{2.641\ \text{g}}{44.01\ \text{g/mol}} = 0.06001\ \text{mol}$
>
> $n(H_2O) = \dfrac{1.351\ \text{g}}{18.02\ \text{g/mol}} = 0.07497\ \text{mol}$, so $n(\text{H}) = 2 \times 0.07497\ \text{mol} = 0.1499\ \text{mol}$
>
> Ratio H : C = 0.1499 mol : 0.06001 mol = 2.498 : 1 ≈ 2.5 : 1 = 5 : 2, so the empirical formula is C_2H_5 (29.06 g/mol).
>
> $\dfrac{58.1\ \text{g/mol}}{29.06\ \text{g/mol}} = 2.00$, so the molecular formula is C_4H_{10}.
>
> (b) **butane** and **2-methylpropane**.

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